Lab 7 — Image Filters

Clean combined notebook for **Lab 7(A): Laplacian Filter** and **Lab 7(B): Mathematical and Statistical Filters**.

Image files required: `../images/moon.jpg` and `../images/noise.jpg` (relative to notebook location).

Lab 7(A) - Laplacian Filter

In [1]:
```
import cv2
import numpy as np

from IPython.display import Image, display
from matplotlib import pyplot as plt

plt.rc('text', usetex=False)
```

Apply Laplacian Filter on Image to get Sharp Image

In [2]:
```
# ================ ( Laplacian Filter ) ==================


# 0. Image file path
imgFile = "../images/moon.jpg"

# 1. Load image and convert to grayscale
img = cv2.imread(imgFile, 0)

# 2. Manually define the Laplacian Kernel
# This is the standard 3x3 kernel that detects horizontal and vertical edges
manual_kernel = np.array([
    [0,  1,  0],
    [1, -4,  1],
    [0,  1,  0]
], dtype=np.float32)

# Alternative kernel: Detects diagonal edges as well
# manual_kernel = np.array([
#     [1,  1,  1],
#     [1, -8,  1],
#     [1,  1,  1]
# ], dtype=np.float32)

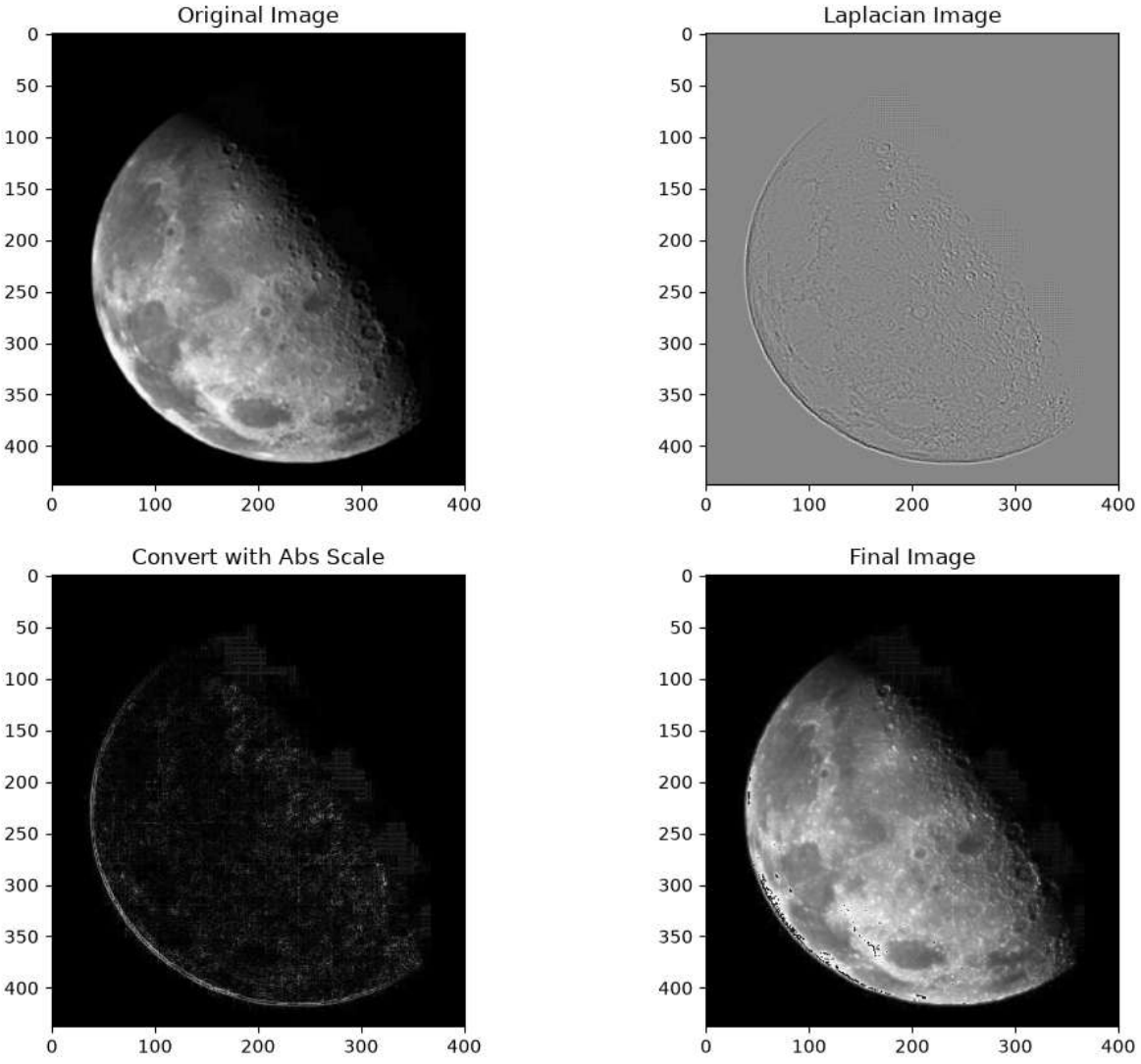
# 3. Apply the filter using filter2D
# -1 indicates the output image will have the same depth as the source
# However, for Laplacian, use cv2.CV_64F to avoid clipping negative values
laplacian_manual = cv2.filter2D(img, cv2.CV_64F, manual_kernel)

# 4. Take absolute value and convert back to 8-bit (0-255)
```

```
laplacian_abs = cv2.convertScaleAbs(laplacian_manual)

finalImage = img + laplacian_abs

# 5. Display images
plt.figure(figsize=(12, 10))
plt.subplot(221), plt.imshow(img, cmap='gray'), plt.title('Original Image')
plt.subplot(222), plt.imshow(laplacian_manual, cmap='gray'), plt.title('Laplacia
plt.subplot(223), plt.imshow(laplacian_abs, cmap='gray'), plt.title('Convert wit
plt.subplot(224), plt.imshow(finalImage, cmap='gray'), plt.title('Final Image')
plt.show()
```



In [3]:
```
# ================== ( Laplacian Filter ) ===============

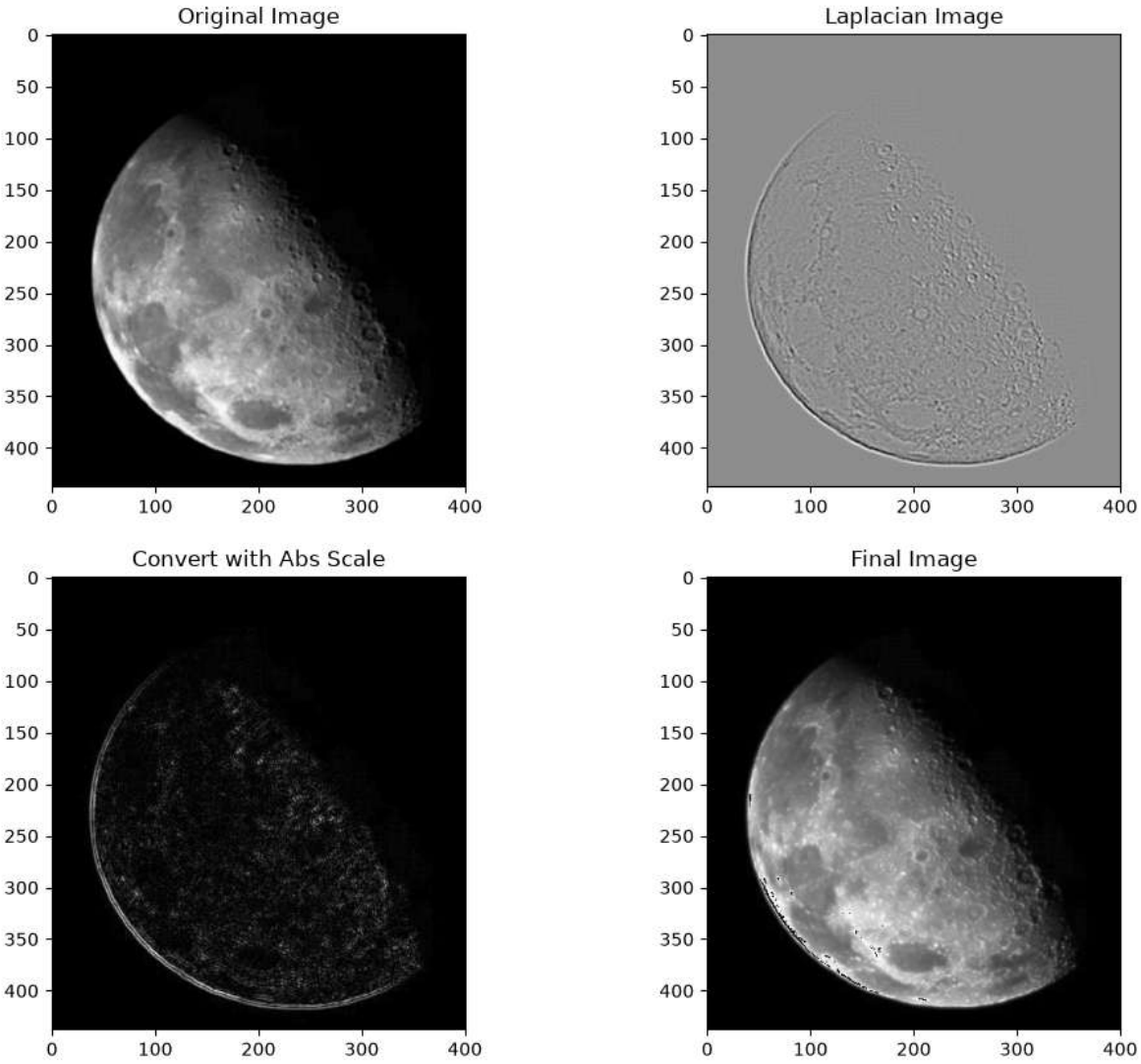
# 1. Optional: Apply Gaussian Blur to reduce noise
# This prevents the algorithm from picking up tiny details as edges
blurred = cv2.GaussianBlur(img, (3, 3), 0)

# 2. Apply the Laplacian Algorithm
# Use CV_64F to capture both positive and negative gradients (essential for edge
laplacian = cv2.Laplacian(blurred, cv2.CV_64F)

# 3. Convert back to 8-bit image to view it properly
laplacian_8bit = cv2.convertScaleAbs(laplacian)

# 4. Adding Laplace and origianl Image
finalImage2 = img + laplacian_8bit
```

```
# 5. Display images
plt.figure(figsize=(12, 10))
plt.subplot(221), plt.imshow(img, cmap='gray'), plt.title('Original Image')
plt.subplot(222), plt.imshow(laplacian, cmap='gray'), plt.title('Laplacian Image
plt.subplot(223), plt.imshow(laplacian_8bit, cmap='gray'), plt.title('Convert wi
plt.subplot(224), plt.imshow(finalImage2, cmap='gray'), plt.title('Final Image')
plt.show()
```



Lab 7(B) - Mathematical and Statistical Filters

In [4]:
```
import cv2
import numpy as np

from IPython.display import Image, display
from matplotlib import pyplot as plt

plt.rc('text', usetex=False)
```

In [5]:
```
# ========== ( Adding nosie to an Image ) =================

# 0. Read an Image
imgPath = '../images/noise.jpg'
```

```
img = cv2.imread(imgPath, 0)

# 1. Function that handle all noise algorithm (Guassian, Exp, Uniform & Salt&Pep
def add_noise(image, noise_type):
    row, col = image.shape

    if noise_type == "gaussian":
        mean = 0
        var = 0.01  # Adjust for more/less noise
        sigma = var**0.5
        gauss = np.random.normal(mean, sigma, (row, col))
        noisy = image / 255 + gauss
        return np.clip(noisy * 255, 0, 255).astype(np.uint8)

    elif noise_type == "salt_pepper":
        s_vs_p = 0.5
        amount = 0.04
        noisy = np.copy(image)
        # Salt (white pixels)
        num_salt = np.ceil(amount * image.size * s_vs_p)
        coords = [np.random.randint(0, i - 1, int(num_salt)) for i in image.shap
        noisy[tuple(coords)] = 255
        # Pepper (black pixels)
        num_pepper = np.ceil(amount * image.size * (1.0 - s_vs_p))
        coords = [np.random.randint(0, i - 1, int(num_pepper)) for i in image.sh
        noisy[tuple(coords)] = 0
        return noisy

    elif noise_type == "exponential":
        # Scale is the inverse of the rate (lambda)
        expo = np.random.exponential(scale=10.0, size=(row, col))
        noisy = image + expo
        return np.clip(noisy, 0, 255).astype(np.uint8)

    elif noise_type == "uniform":
        low = -20
        high = 20
        uni = np.random.uniform(low, high, (row, col))
        noisy = image + uni
        return np.clip(noisy, 0, 255).astype(np.uint8)



# 2. Load image in grayscale
img = cv2.imread(imgPath, 0)

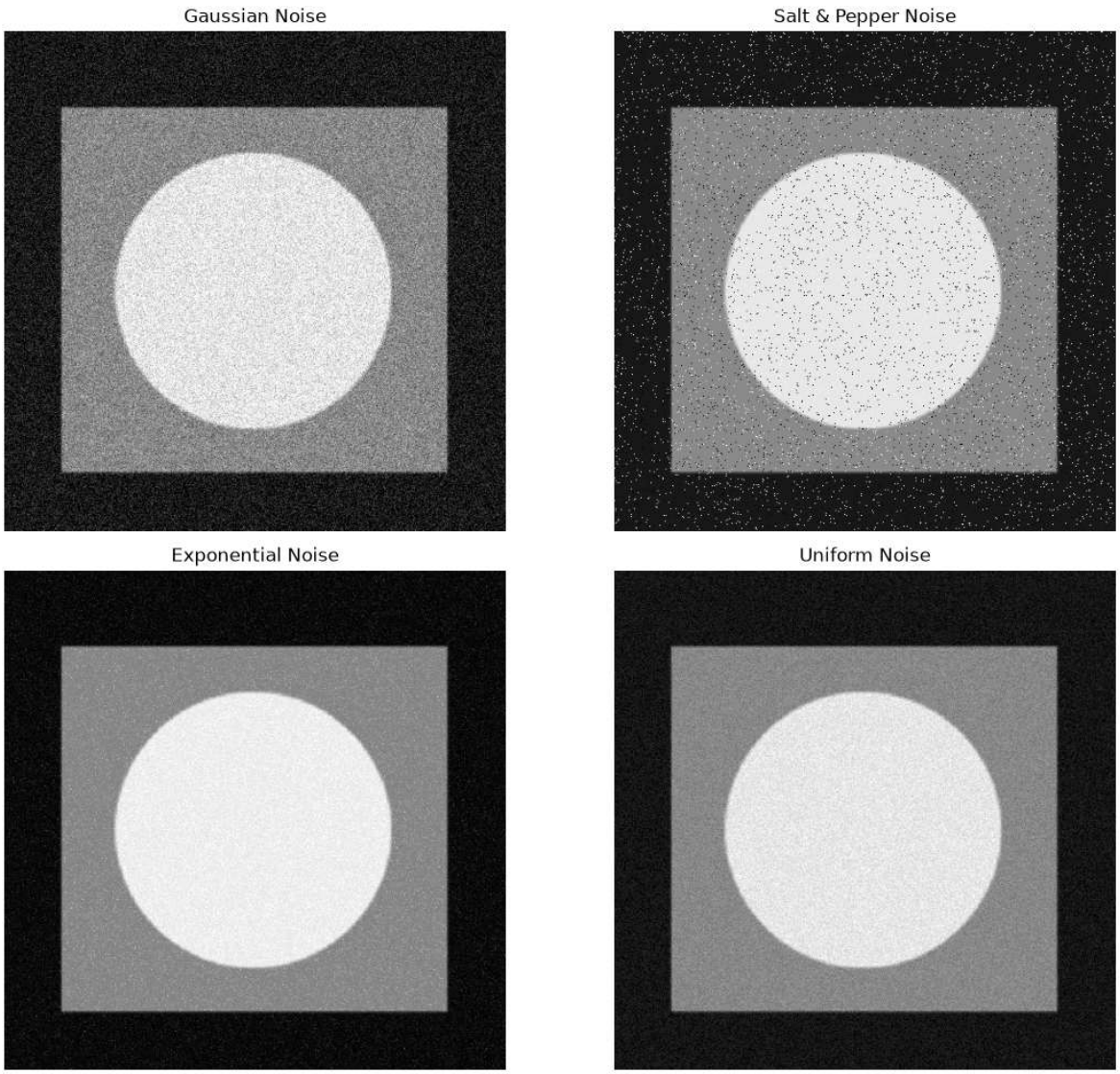
# 3. Generate Noisy Images
noise_gauss = add_noise(img, "gaussian")
noise_sp = add_noise(img, "salt_pepper")
noise_expo = add_noise(img, "exponential")
noise_uni = add_noise(img, "uniform")

# 4. Display in 2x2 Grid
plt.figure(figsize=(12, 10))

plt.subplot(221), plt.imshow(noise_gauss, cmap='gray'), plt.title('Gaussian Nois
plt.subplot(222), plt.imshow(noise_sp, cmap='gray'), plt.title('Salt & Pepper No
plt.subplot(223), plt.imshow(noise_expo, cmap='gray'), plt.title('Exponential No
plt.subplot(224), plt.imshow(noise_uni, cmap='gray'), plt.title('Uniform Noise')
```

```
for i in range(1, 5):
    plt.subplot(220 + i).axis('off')

plt.tight_layout()
plt.show()
```



In [6]:
```
# --- METHOD 1: Using Built-in OpenCV function ---

# 2. Apply Arithmetic Mean Filter (using a 5x5 blur window)
# The larger the numbers in (3, 3), the smoother the image becomes.,
# you can change kernel size to (5,5), (7,7) etc.
filtered_img = cv2.blur(noise_gauss, (3, 3))

display(filtered_img)
```

```
array([[12, 23, 14, ..., 14, 18, 23],
       [26, 31, 25, ..., 25, 27, 28],
       [28, 31, 29, ..., 20, 22, 26],
       ...,
       [37, 32, 33, ..., 27, 24, 29],
       [41, 32, 32, ..., 24, 30, 41],
       [39, 36, 37, ..., 21, 25, 30]], shape=(506, 508), dtype=uint8)
```

In [7]:
```
# --- METHOD 2: Manual implementation using a Kernel ---

# An Arithmetic Mean kernel is just a matrix of 1s divided by the total number o
```

```
kernel_size = 3
kernel = np.ones((kernel_size, kernel_size), np.float32) / (kernel_size * kernel
arithmetic_mean_manual = cv2.filter2D(img, -1, kernel)

display(arithmetic_mean_manual)
```

```
array([[24, 24, 24, ..., 24, 24, 24],
       [24, 24, 24, ..., 24, 24, 24],
       [24, 24, 24, ..., 24, 24, 24],
       ...,
       [24, 24, 24, ..., 24, 24, 24],
       [24, 24, 24, ..., 24, 24, 24],
       [24, 24, 24, ..., 24, 24, 24]], shape=(506, 508), dtype=uint8)
```

Geometric Means

In [8]:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

def geometric_mean_filter(image, kernel_size=3):
    # 1. Convert to float64 and add a tiny epsilon to avoid log(0)
    img_float = image.astype(np.float64) + 1e-6

    # 2. Apply the log-transform
    log_img = np.log(img_float)

    # 3. Apply a box filter to the log image (calculates the average of logs)
    kernel = np.ones((kernel_size, kernel_size), np.float32) / (kernel_size * ke
    mean_log = cv2.filter2D(log_img, -1, kernel)

    # 4. Take the exponent to get back to the geometric mean
    geo_mean = np.exp(mean_log)

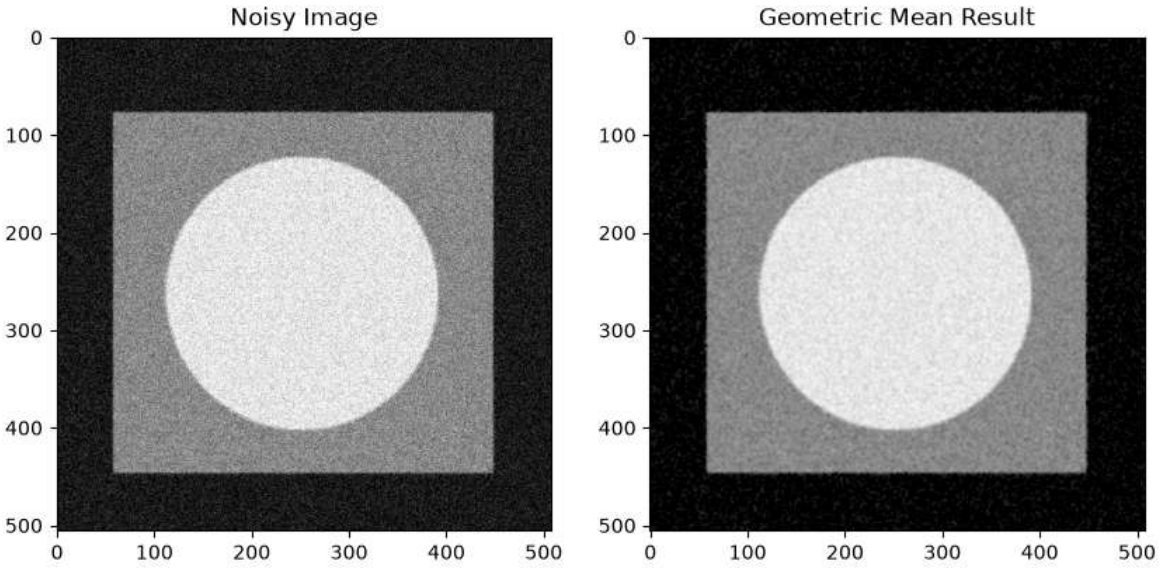
    # 5. Clip and convert back to 8-bit
    return np.clip(geo_mean, 0, 255).astype(np.uint8)


# === List of noisy images
# noise_gauss = add_noise(img, "gaussian")
# noise_sp = add_noise(img, "salt_pepper")
# noise_expo = add_noise(img, "exponential")
# noise_uni = add_noise(img, "uniform")

noise_img = noise_gauss

# Apply filter
result = geometric_mean_filter(noise_img, kernel_size=3)

# Display Comparison
plt.figure(figsize=(10, 5))
plt.subplot(121), plt.imshow(noise_img, cmap='gray'), plt.title('Noisy Image')
plt.subplot(122), plt.imshow(result, cmap='gray'), plt.title('Geometric Mean Res
plt.show()
```



Harmonic Means Filter

In [9]:
```
def harmonic_mean_filter(image, kernel_size=3):
    # 1. Convert to float64 to avoid overflow during calculation
    # Add a tiny epsilon to handle black pixels (0)
    img_float = image.astype(np.float64) + 1e-6

    # 2. Calculate the reciprocal of every pixel (1/P)
    reciprocal_img = 1.0 / img_float

    # 3. Sum the reciprocals in a window and divide by window size
    # This is equivalent to applying a standard mean filter to the reciprocal im
    kernel = np.ones((kernel_size, kernel_size), np.float32)
    sum_reciprocal = cv2.filter2D(reciprocal_img, -1, kernel)

    # 4. Final step of the formula: (Window Size) / Sum of Reciprocals
    # Note: Window size is kernel_size * kernel_size
    n = kernel_size * kernel_size
    harmonic_mean = n / sum_reciprocal

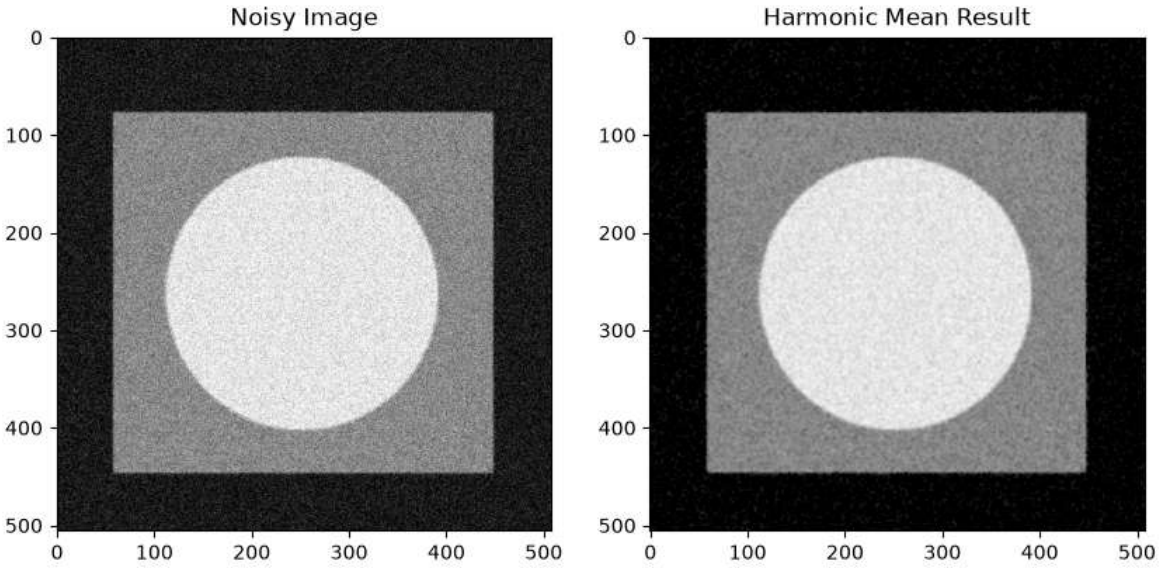
    # 5. Clip the results and convert back to 8-bit
    return np.clip(harmonic_mean, 0, 255).astype(np.uint8)


# === List of noisy images
# noise_gauss = add_noise(img, "gaussian")
# noise_sp = add_noise(img, "salt_pepper")
# noise_expo = add_noise(img, "exponential")
# noise_uni = add_noise(img, "uniform")

noise_img = noise_gauss

# Apply the filter
cleaned_img = harmonic_mean_filter(noise_img, kernel_size=3)

# Display Comparison
plt.figure(figsize=(10, 5))
plt.subplot(121), plt.imshow(noise_img, cmap='gray'), plt.title('Noisy Image')
plt.subplot(122), plt.imshow(cleaned_img, cmap='gray'), plt.title('Harmonic Mean
plt.show()
```



Contra - Harmonic Mean Filter

In [10]:
```
def contra_harmonic_mean(image, kernel_size=3, Q=0):
    # 1. Convert to float to handle high powers and prevent overflow
    img_float = image.astype(np.float64) + 1e-6

    # 2. Calculate numerator: Sum of pixels to the power (Q + 1)
    numerator = np.power(img_float, Q + 1)
    kernel = np.ones((kernel_size, kernel_size), np.float32)
    num_sum = cv2.filter2D(numerator, -1, kernel)

    # 3. Calculate denominator: Sum of pixels to the power (Q)
    denominator = np.power(img_float, Q)
    den_sum = cv2.filter2D(denominator, -1, kernel)

    # 4. Divide and clean up
    # Handle potential division by zero
    result = num_sum / (den_sum + 1e-6)

    return np.clip(result, 0, 255).astype(np.uint8)

# === List of noisy images
# noise_gauss = add_noise(img, "gaussian")
# noise_sp = add_noise(img, "salt_pepper")
# noise_expo = add_noise(img, "exponential")
# noise_uni = add_noise(img, "uniform")

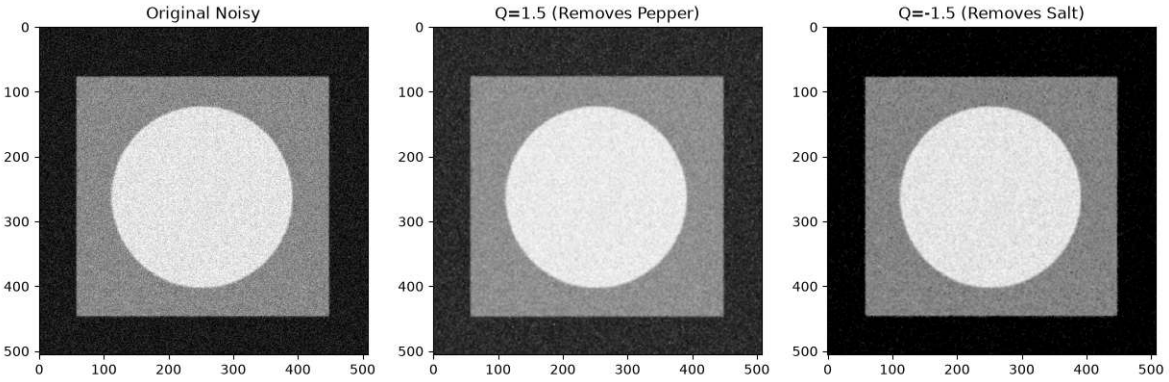
noise_img = noise_gauss

# --- APPLICATIONS ---
# Q > 0 (e.g., 1.5): Removes Pepper Noise (black spots)
pepper_removed = contra_harmonic_mean(noise_img, kernel_size=3, Q=1.5)

# Q < 0 (e.g., -1.5): Removes Salt Noise (white spots)
salt_removed = contra_harmonic_mean(noise_img, kernel_size=3, Q=-1.5)

# --- DISPLAY ---
plt.figure(figsize=(15, 5))
plt.subplot(131), plt.imshow(noise_img, cmap='gray'), plt.title('Original Noisy'
plt.subplot(132), plt.imshow(pepper_removed, cmap='gray'), plt.title('Q=1.5 (Rem
```

```
plt.subplot(133), plt.imshow(salt_removed, cmap='gray'), plt.title('Q=-1.5 (Remc
plt.show()
```



Median Filter

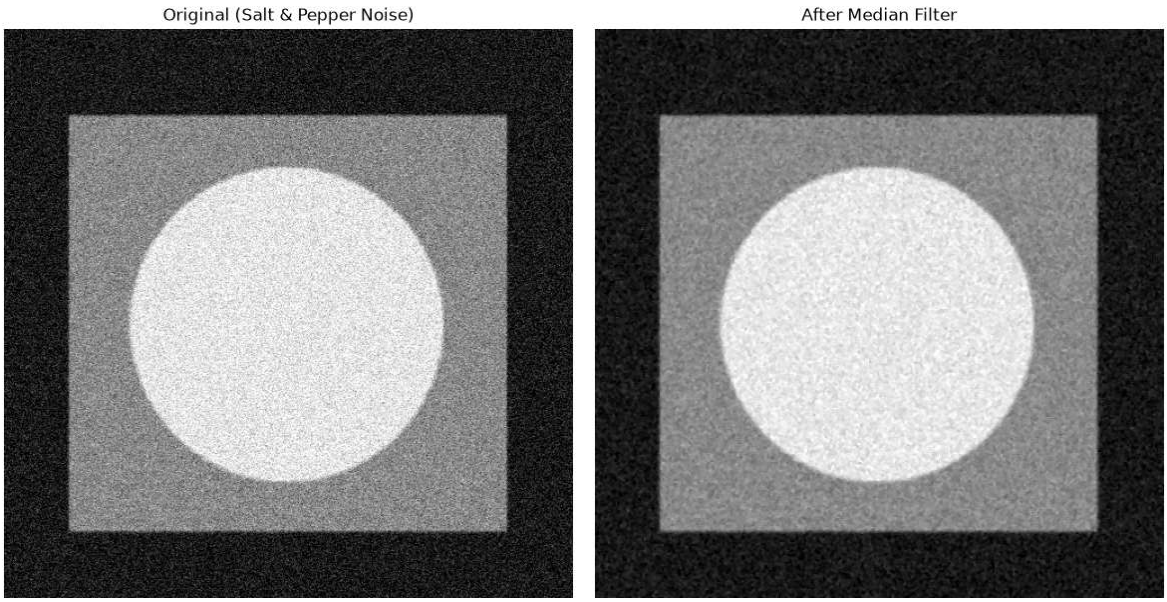
In [11]:
```
# === List of noisy images
# noise_gauss = add_noise(img, "gaussian")
# noise_sp = add_noise(img, "salt_pepper")
# noise_expo = add_noise(img, "exponential")
# noise_uni = add_noise(img, "uniform")

noise_img = noise_gauss

# 2. Apply Median Filter
# The second argument is the 'ksize' (kernel size).
# It MUST be an odd integer (3, 5, 7, etc.)
median_result = cv2.medianBlur(noise_img, 3)

# 3. Compare Results
plt.figure(figsize=(12, 6))

plt.subplot(121), plt.imshow(noise_img, cmap='gray'), plt.title('Original (Salt
plt.subplot(122), plt.imshow(median_result, cmap='gray'), plt.title('After Media
plt.tight_layout()
plt.show()
```



Max and Min Filter

In [12]:
```
# === List of noisy images
# noise_gauss = add_noise(img, "gaussian")
# noise_sp = add_noise(img, "salt_pepper")
# noise_expo = add_noise(img, "exponential")
# noise_uni = add_noise(img, "uniform")

noise_img = noise_gauss



# 2. Define the Kernel size (e.g., 3x3)
kernel = np.ones((3, 3), np.uint8)

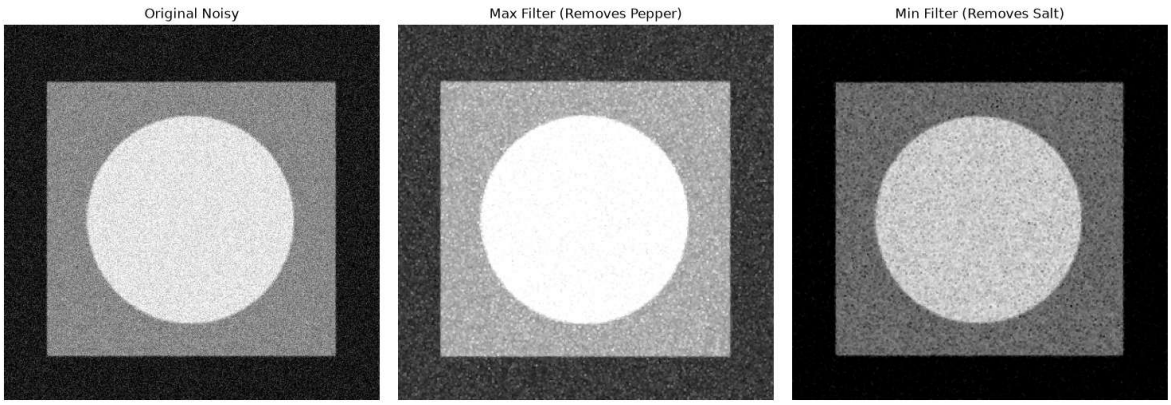
# 3. Apply Max Filter (Dilation)
# This picks the brightest pixel in the neighborhood.
max_filter = cv2.dilate(noise_img, kernel)

# 4. Apply Min Filter (Erosion)
# This picks the darkest pixel in the neighborhood.
min_filter = cv2.erode(noise_img, kernel)

# --- Display Results ---
plt.figure(figsize=(15, 5))

plt.subplot(131), plt.imshow(noise_img, cmap='gray'), plt.title('Original Noisy'
plt.subplot(132), plt.imshow(max_filter, cmap='gray'), plt.title('Max Filter (Re
plt.subplot(133), plt.imshow(min_filter, cmap='gray'), plt.title('Min Filter (Re

for ax in plt.gcf().axes: ax.axis('off')
plt.tight_layout()
plt.show()
```



Mid-Point Filtering

In [13]:
```
def midpoint_filter(image, kernel_size=3):
    # 1. Define the structural element (kernel)
    kernel = np.ones((kernel_size, kernel_size), np.uint8)

    # 2. Find the Maximum value in the neighborhood (Dilation)
    max_img = cv2.dilate(image, kernel).astype(np.float64)

    # 3. Find the Minimum value in the neighborhood (Erosion)
    min_img = cv2.erode(image, kernel).astype(np.float64)
```

```
    # 4. Calculate the average of Max and Min
    midpoint = (max_img + min_img) / 2

    # 5. Convert back to 8-bit
    return midpoint.astype(np.uint8)



# === List of noisy images
# noise_gauss = add_noise(img, "gaussian")
# noise_sp = add_noise(img, "salt_pepper")
# noise_expo = add_noise(img, "exponential")
# noise_uni = add_noise(img, "uniform")

noise_img = noise_gauss


# Apply the Midpoint Filter
result = midpoint_filter(noise_img, kernel_size=3)

# --- Display Results ---
plt.figure(figsize=(12, 6))

plt.subplot(121), plt.imshow(noise_img, cmap='gray'), plt.title('Original (Gauss
plt.subplot(122), plt.imshow(result, cmap='gray'), plt.title('After Midpoint Fil

plt.tight_layout()
plt.show()
```

